Akoizo Original Practice - Form 5 - Answer key

Division B | Hovercraft | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) How may rolls of nickels be attached to the vehicle?

Unit: Carrying nickels: mass, balance, and security.
Worked-case data: A vehicle carries 3 rolls, the half roll, and 12 loose nickels, but 2 loose nickels fall out mid-run. The run is complete. Find the Mass Score and the penalty effect.
Separate investigation observations:
Run 1: 4 rolls in cradles, 20 loose in an open tray: 3 nickels lost at start.
Run 2: Same load, tray with 2 cm walls: nothing lost.
Run 1 score effect: Penalty x 0.8 applied to the whole run.

A. Held in a shaped cradle without tape or adhesive

B. Taped to the deck

C. Unrolled into a tray

D. Glued in place

Correct: A

2. (2 points) Why can starting farther behind the start line shorten the timed run?

A. The vehicle builds speed before timing begins

B. The track is shorter

C. The supervisor starts timing later

D. It reduces vehicle mass

Correct: A

3. (2 points) A team uses the ramp and finishes exactly on TT carrying 2 rolls. What is the run score?

A. 60

B. 100

C. 80

D. 48

Correct: A

4. (2 points) Why are papers with calibration data impounded with the vehicle?

Unit: Hitting the Target Time: calibration and repeatability.
Worked-case data: Your curve shows setting 3 -> 9.6 s and setting 4 -> 11.2 s (full load, 5.0 V). The Target Time is 10.0 s. Which setting do you choose?
Separate investigation observations:
Calibration: Settings 2, 4, 6 gave 8.0, 12.0, 16.0 s.
Target Time: 13.0 s announced.
Run 1 at 4.5: 13.5 s.

A. They replace the run

B. They are a tiebreaker

C. No additional notes may be brought in after impound

D. The supervisor grades them

Correct: C

5. (4 points) Why is a microcontroller timer not an option in this event?

Unit: The electrical system: batteries, motors, and speed control.
Worked-case data: A team wants to use eight 1.5 V alkaline AA cells. Is that legal, and what are the options?
Separate investigation observations:
Monday: Fresh pack 5.2 V, knob at 3: 11.8 s.
Wednesday: Same knob, pack at 4.7 V: 13.4 s.
Wednesday, recharged: Pack 5.2 V, knob at 3: 11.9 s.

answer: Integrated circuits are not permitted except as part of a commercial motor, so timing must come from analog components, mechanical systems, or calibration. (4 points).

6. (4 points) Why does a larger footprint reduce the fan power needed?

Unit: Levitation: the air cushion and the skirt.
Worked-case data: A hovercraft with nickels weighs 1.6 kg and has a 0.35 m x 0.35 m cushion. What cushion pressure is needed?
Separate investigation observations:
Empty test: Vehicle glides freely across the floor with lift on.
With all nickels: Rear skirt drags and the craft stops quickly.
Nickels moved forward 6 cm: Glides freely again, sitting level.

answer: It lowers the required cushion pressure, and leakage through the hover gap drops at lower pressure, so the fan moves less air against less pressure. (4 points).

7. (4 points) What should a team do if inspection finds the dowel 4 cm from the front?

answer: Move it to within 3.0 cm of the front edge during the testing period, keeping it at least 20 cm above the track. (4 points).

8. (4 points) Explain how an off-center load can slow the vehicle.

Unit: Carrying nickels: mass, balance, and security.
Worked-case data: A vehicle carries 3 rolls, the half roll, and 12 loose nickels, but 2 loose nickels fall out mid-run. The run is complete. Find the Mass Score and the penalty effect.
Separate investigation observations:
Run 1: 4 rolls in cradles, 20 loose in an open tray: 3 nickels lost at start.
Run 2: Same load, tray with 2 cm walls: nothing lost.
Run 1 score effect: Penalty x 0.8 applied to the whole run.

answer: It tilts the hull, raising the gap on one side so air escapes and pressing the skirt into the floor on the other, adding friction. (4 points).

9. (4 points) Why is a vehicle that barely reaches the finish risky?

Unit: Thrust, friction, and motion down the track.
Worked-case data: A hovercraft must cross 185 cm in 12.0 s at steady speed. It enters the timed zone already at speed. What speed is required, and how does adding 0.2 kg affect it?
Separate investigation observations:
Trial 1: Start 5 cm behind line: 14.1 s.
Trial 2: Start 40 cm behind line: 11.6 s.
Trial 3: Start 40 cm behind, plus 2 rolls: 13.8 s.

answer: If it slows almost to a stop it can stall for 3 seconds or exceed 2 x TT, making the run incomplete with no Time Score. (4 points).

10. (6 points) Solve this case and explain your reasoning: TT is 12.0 s. A vehicle carries 3 rolls and 10 loose nickels and finishes in 13.2 s with nothing falling off. Find the run score.

reasoning: DS = 40 (complete run).
TS = 40 x (1 - 1.2/12.0) = 36.00.
MS = (4 x 3 + 0.1 x 10) x 185/185 = 13. (3 points).

conclusion: Run score = 40 + 36.00 + 13 = 89.00. (3 points).

11. (6 points) Solve this case and explain your reasoning: Your curve shows setting 3 -> 9.6 s and setting 4 -> 11.2 s (full load, 5.0 V). The Target Time is 10.0 s. Which setting do you choose?

reasoning: The target is (10.0 - 9.6) ÷ (11.2 - 9.6) = 0.25 of the way from 3 to 4.
Interpolated setting = 3 + 0.25 = 3.25.
If the first run gives 10.3 s, TS = 38.80; shift slightly toward 3 for run two. (3 points).

conclusion: Interpolation gives a starting setting; the first run tells you which way to correct. (3 points).

12. (2 points) Why did Wednesday's first run take longer?

Supplied synthetic observations:
Monday: Fresh pack 5.2 V, knob at 3: 11.8 s.
Wednesday: Same knob, pack at 4.7 V: 13.4 s.
Wednesday, recharged: Pack 5.2 V, knob at 3: 11.9 s.

A. The pack voltage was lower

B. The knob drifted to a new setting

C. The track was longer

Correct: A

13. (2 points) Define "Cushion pressure" in the context of Hovercraft.

answer: Pressure above atmospheric under the hull; equals weight ÷ area. (2 points).

14. (2 points) Define "Impound" in the context of Hovercraft.

answer: Turning in the vehicle, spares, and papers before competition. (2 points).

15. (2 points) Define "Half roll" in the context of Hovercraft.

answer: 20 nickels, counts as R = 0.5. (2 points).

16. (2 points) Define "Acceleration" in the context of Hovercraft.

answer: Rate of change of velocity, a = F_net/m. (2 points).

17. (2 points) Define "Mass Score (MS)" in the context of Hovercraft.

answer: (4R + 0.1N) prorated by distance; up to 20. (2 points).

18. (2 points) Define "Accuracy" in the context of Hovercraft.

answer: How close a result is to the target. (2 points).